

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 6: Absorption, Respiration and Regulation: Explaining the Fall Back to 4.5 mmol/L

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
B	B	B	A	C	D	C	A

USE AFTER: PREDICT PHASE — Predict: 'What would the graph look like if nothing removed glucose from the blood?' (dotted line sketch)

Q1. Wanjiru's blood glucose reached a peak of 8.0 mmol/L at 60 minutes after her meal of ugali, sukuma wiki and beans. Suppose nothing removed glucose from her blood after that time. What would her blood glucose concentration do?

- A. It would keep rising without limit because glucose is made in the blood
- B. It would stay high, close to 8.0 mmol/L, because nothing is taking glucose out**
- C. It would fall back to 4.5 mmol/L anyway because glucose slowly disappears
- D. It would fall because glucose is lost in sweat and urine

Correct answer: B. The fall on the graph is caused by processes that remove glucose (uptake by cells, respiration, storage). If nothing removed it, the level would stay high. Glucose does not simply disappear or get lost in sweat.

USE AFTER: OBSERVE PHASE — Part 1: Data analysis in groups of 4 (calculating the fall in each interval)

Q2. Wanjiru's blood glucose was 8.0 mmol/L at 60 minutes and 6.5 mmol/L at 90 minutes. By how much did it fall in this 30-minute interval?

- A. 3.5 mmol/L
- B. 1.5 mmol/L**
- C. 0.5 mmol/L
- D. 14.5 mmol/L

Correct answer: B. Fall = $8.0 - 6.5 = 1.5$ mmol/L. 3.5 mmol/L is the total fall over 120 minutes, not this one interval.

USE AFTER: OBSERVE PHASE — Part 1: Data analysis in groups of 4 (comparing the rise with the fall)

Q3. Wanjiru's blood glucose rose by 3.5 mmol/L in the first 60 minutes (4.5 to 8.0). It then fell by 3.5 mmol/L from 60 to 180 minutes (8.0 to 4.5). How many times longer did the fall take than the rise?

- A. 3.5 times as long
- B. 2 times as long**
- C. The same time, because both changes are 3.5 mmol/L
- D. Half as long, because the body removes glucose quickly

Correct answer: B. The rise took 60 min and the fall took $180 - 60 = 120$ min, so the fall took $120 \div 60 = 2$ times as long. The change in concentration is the same but the fall is slower.

USE AFTER: OBSERVE PHASE — Part 2: Role-play 'Journey of the glucose' (pancreas holds insulin cards)

Q4. In the role-play of glucose after a meal, a learner holds up insulin cards when the blood glucose is high. Which statement correctly describes insulin?

- A. The pancreas releases insulin, and it causes cells, the liver and muscles to take up glucose from the blood**
- B. The small intestine releases insulin, and it moves glucose into the blood
- C. Insulin is a food that gives the body energy when we eat
- D. The liver releases insulin, and it digests starch into glucose

Correct answer: A. Insulin is a hormone made by the pancreas. It is released when blood glucose is high and causes cells, liver and muscles to take up glucose, so blood glucose falls.

USE AFTER: OBSERVE PHASE — Part 2: Role-play 'Journey of the glucose' (step 5: respiration in the cells)

Q5. Inside body cells, glucose is used in respiration to release energy. Which equation correctly shows respiration?

- A. Glucose + water gives starch + oxygen + energy
- B. Glucose + carbon dioxide gives oxygen + water + energy
- C. Glucose + oxygen gives carbon dioxide + water + energy**
- D. Glucose + oxygen gives glycogen + carbon dioxide + energy

Correct answer: C. In respiration, glucose reacts with oxygen to produce carbon dioxide and water, and energy (ATP) is released. This is also why we breathe harder after running to catch a matatu.

USE AFTER: OBSERVE PHASE — Part 2: Role-play 'Journey of the glucose' (step 6: liver and muscles store glycogen)

Q6. After a meal, some glucose is not needed straight away. What happens to this excess glucose?

- A. It is removed from the body as sugar in urine
- B. It is changed to insulin in the pancreas
- C. It is destroyed in the blood
- D. It is stored as glycogen in the liver and muscles**

Correct answer: D. The liver and muscles store excess glucose as glycogen. This helps to bring blood glucose back to normal. It is not simply lost in urine or destroyed.

USE AFTER: EXPLAIN PHASE — Final Explanation (Claim-Evidence-Reasoning): choosing evidence that glucose came from the starch in ugali

Q7. Learners write a Claim-Evidence-Reasoning explanation of Wanjiru's graph. Which piece of evidence best shows that the glucose in the blood came from digesting the starch in ugali?

- A. Her blood glucose fell by 3.5 mmol/L over 120 minutes
- B. The normal fasting range of blood glucose is 4.0 to 5.5 mmol/L
- C. Saliva containing amylase, kept at 37 degrees C with starch, gave a brick-red colour with Benedict's test after 10 minutes**
- D. Sukuma wiki is a green vegetable

Correct answer: C. The brick-red colour shows that enzyme action turned starch into a reducing sugar. This links starch to glucose. The other choices do not show where the glucose came from.

Q8. Which flow diagram shows the correct order of the final model that explains Wanjiru's blood glucose graph?

- A. Food chemicals, then digestion by enzymes, then glucose in blood, then cells (respiration) and storage, then back to normal**
- B. Food chemicals, then glucose in blood, then digestion by enzymes, then cells and storage, then back to normal
- C. Digestion by enzymes, then food chemicals, then glucose in blood, then back to normal, then cells and storage
- D. Food chemicals, then cells and storage, then digestion by enzymes, then glucose in blood, then back to normal

Correct answer: A. Food is first digested by enzymes into glucose, which is absorbed into the blood. Cells use it in respiration and the liver and muscles store it, so the level returns to normal (4.5 mmol/L).